Math 4141: Geometry and Differential Calculus

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Math 4141: Geometry and Differential Calculus **Credit: 4.00**

Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Exercise

Q. Find the value of λ if the following equations may represent pairs of straight lines $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$.

Solution: Given $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$

Now compare with equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = 12, h = -5, b = 2, g = \frac{11}{2}, f = -\frac{5}{2}$ and $c = \lambda$

We have $\Delta = 0$
Find the value of λ yourself.

Q. Find the value of λ if the following equations may represent pairs of straight lines $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$.

Reduce the equation in standard form

Q. Prove that $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ represents a parabola. Also find the focus and directrix.

Solution

Given that

$$9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$$

Compare the given equation with the general second-degree equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0.$$

We obtain

$$a = 9, h = -12, b = 16, g = -10, f = -\frac{15}{2}, c = -60.$$

First,

$$h^2 - ab = (-12)^2 - (9)(16) = 144 - 144 = 0.$$

The determinant of the conic is

$$\Delta = \begin{vmatrix} 9 & -12 & -10 \\ -12 & 16 & -\frac{15}{2} \\ -10 & -\frac{15}{2} & -60 \end{vmatrix} = -\frac{15625}{4} \neq 0.$$

Thus, the conic is a parabola.

Now rearrange the equation:

$$9x^2 - 24xy + 16y^2 = 20x + 15y + 60,$$
$$(3x - 4y)^2 = 5(4x + 3y + 12).$$

The lines $3x - 4y = 0$ and $4x + 3y + 12 = 0$ are perpendicular.

Their slopes are $\frac{3}{4}$ and $-\frac{4}{3}$, and their product is -1.

Introduce coordinates along these perpendicular directions:

$$Y = \frac{3x - 4y}{\sqrt{3^2 + (-4)^2}} = \frac{3x - 4y}{5}, \qquad X = \frac{4x + 3y + 12}{\sqrt{4^2 + 3^2}} = \frac{4x + 3y + 12}{5}.$$

Since $3x - 4y = 5Y$ and $4x + 3y + 12 = 5X$, substitution gives

$$(5Y)^2 = 5(5X) \implies Y^2 = X.$$

Compare this with the standard form $Y^2 = 4aX$. Therefore,

$$4a = 1 \implies a = \frac{1}{4}.$$

The parabola opens in the positive X-direction.

Reduce the equation in standard form – Solution Continues

Q. Prove that $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ represents a parabola. Also find the focus and directrix.

Focus

For $Y^2 = 4aX$, the focus in the (X, Y) coordinate system is $(a, 0)$. Here it is $\left(\frac{1}{4}, 0\right)$. Using the coordinate definitions,

$$\frac{3x - 4y}{5} = 0, \qquad \frac{4x + 3y + 12}{5} = \frac{1}{4}.$$

Thus,

$$3x - 4y = 0, \qquad 4x + 3y = -\frac{43}{4}.$$

Solving these simultaneous equations gives

$$\boxed{\text{Focus} = \left(-\frac{43}{25}, -\frac{129}{100}\right)}.$$

Directrix

For $Y^2 = 4aX$, the directrix is $X = -a$. Therefore $X = -\frac{1}{4}$, so

$$\frac{4x + 3y + 12}{5} = -\frac{1}{4},$$
$$4x + 3y + 12 = -\frac{5}{4}.$$

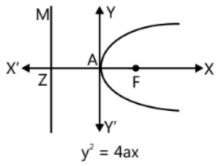
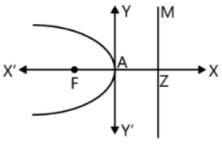
Therefore, the directrix is

$$\boxed{16x + 12y + 53 = 0}.$$

Hence, the given equation represents a non-degenerate parabola with focus $\left(-\frac{43}{25}, -\frac{129}{100}\right)$ and directrix $16x + 12y + 53 = 0$.

Supporting material

3. STANDARD FORMS OF PARABOLA

Right handed parabola: The equation of this type of parabola is of the form $y^2 = 4ax$, $a > 0$. See Fig. 10.7	 $y^2 = 4ax$ **Figure 10.7**
Left handed parabola: The equation of this type of parabola is of the form $y^2 = -4ax$, $a > 0$. See Fig. 10.8	 $y^2 = -4ax$ **Figure 10.8**